

JULIAN MONZAVIFAR

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EDUCATION

North Carolina State University:

B.S. Mechanical Engineering: GPA: 3.86/4.0

Raleigh, NC

Aug 2025 - May 2029

SKILLS & CERTIFICATIONS

- Certified SolidWorks Associate with 5+ years of CAD experience in SolidWorks and Onshape
- CNC mill/lathe, CAM, FEA, design for manufacturability (DFM), TIG welding, mechanical assembly, Excel

ENGINEERING & PROFESSIONAL EXPERIENCE

Weldon Steel

Weldon, NC

Manufacturing Engineer and Operations Intern

May 2026 - Present

- Designed and implemented data collection device to monitor machine utilization while analyzing operational workflows identifying an improvement for a 30% projected increase in plasma cutter efficiency
- Collaborated with machine operators to map 80% of core fabrication workflow and created training materials to standardize manufacturing processes and improve employee onboarding
- Audited financial data to develop prediction models for revenue recognition aiding in planned Cap Ex projects

Pack Motorsports Baja SAE

Raleigh, NC

Drivetrain Engineer and Manufacturing Lead

Aug 2025 - Present

- Direct drivetrain, suspension, and brakes subsystem redesigns to guarantee DFM, coordinate manufacturing timelines, and personally redesign the vehicle's front U-joint axles and hubs to reduce assembly weight by 20%
- Train new members in CNC operation, CAM programming, and manual machining practices
- Programmed and machined three sets of front axle assemblies alongside the steering housing and half the CVT

Center for Additive Manufacturing and Logistics

Raleigh, NC

CNC and Additive Manufacturing Researcher

Oct 2025- Present

- Operate three and five axis CNC mills to machine various metals using Fusion CAM, and post machine additive manufactured components utilizing custom fixture design
- Collaborate with professors and graduate students to refine designs for manufacturability, create technical drawings, and complete machining for research projects on schedule
- Serve as technical representative, showcasing facility to faculty, prospective students, and industry sponsors

Hampton's Mobile Service

Huntersville, NC

Mechanic

May 2025 - Aug 2025

- Diagnosed mechanical and electrical issues, providing accurate service quotes and repair timelines to customers
- Performed routine maintenance and vehicle repairs to all major components in the undercarriage and engine bay

LEADERSHIP & CO-CURRICULAR ACTIVITIES

Labyrinth Robotics, *FIRST* Robotics Competition Team

Morganton, NC

Team Mentor (Formerly Captain & Head Engineer)

May 2025 - Present

- Coordinate and facilitate engineering training in CAD, manufacturing, and DFM to ensure team knowledge transfer
- Audit technical design viability, enforce project scope to meet season deadlines, and oversee inventory and procurement
- Secured \$10,000 in team funding through grant applications, outreach, and local business partnerships

North Carolina School of Science and Mathematics

Morganton, NC

Residential Life Assistant

Aug 2024 - May 2025

- Supported 36 residents through peer guidance, conflict resolution, weekly hall meetings, and 1:1 check-ins
- Cultivated community by planning and executing 4+ monthly hall events while allocating and managing community budget

HONORS & AWARDS

First Place 2026 Baja SAE Carolina Night, NCSU Dean's List, National Merit Commended Scholar, and NCSSM Outstanding Service Award (Completion of 150 hours of Community Service)